Red Hat AI 3

Validated models

Red Hat AI validated models

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Abstract

Learn about the validated models that you can inference serve with Red Hat AI.

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CHAPTER 1. ABOUT RED HAT AI VALIDATED MODELS

Red Hat AI validated models have been tested and verified to work correctly across supported hardware and product configurations. These models are available as Hugging Face downloads, as OCI artifact images, and as modelcar container images. Platform-specific validated models are also available for IBM Spyre on IBM Power and IBM Z systems.

NOTE

If you are using AI Inference with Podman as part of a RHEL AI deployment, use ModelCar container images or Hugging Face models.

If you are using AI Inference as part of an Red Hat OpenShift AI deployment on OpenShift Container Platform, use OCI artifact images.

Red Hat uses GuideLLM for performance benchmarking and Language Model Evaluation Harness for accuracy evaluations.

Explore the Red Hat AI validated models collections on Hugging Face.

IMPORTANT

AMD GPUs support FP8 (W8A8) and GGUF quantization variant models only. For more information, see Supported hardware.

CHAPTER 2. RED HAT AI VALIDATED MODELS - FEBRUARY 2026

The following models, available from RedHat AI on Hugging Face, are validated for use with Red Hat AI Inference.

Table 2.1. Red Hat AI validated models - February 2026 collection

Model	Quantized variants	Hugging Face model cards	Validated on
granite-4.0-h-small	FP8	• FP8	• RHAIIS 3.3 • RHOAI 3.3
granite-4.0-h-tiny	FP8	• FP8	• RHAIIS 3.3 • RHOAI 3.3
Ministral-3-14B-Instruct-2512	None	• Baseline	• RHAIIS 3.3 • RHOAI 3.3
Phi-4-reasoning	FP8	• FP8	• RHAIIS 3.3 • RHOAI 3.3
Qwen3-Next-80B-A3B-Instruct	INT4	• INT4	• RHAIIS 3.3 • RHOAI 3.3
Qwen3-VL-235B-A22B-Instruct-NVFP4	None	• Baseline	• RHAIIS 3.3 • RHOAI 3.3

CHAPTER 3. RED HAT AI VALIDATED MODELS - JANUARY 2026

The following models, available from RedHat AI on Hugging Face , are validated for use with Red Hat AI Inference.

Table 3.1. Red Hat AI validated models - January 2026 collection

Model	Quantized variants	Hugging Face model cards	Validated on
Apertus-8B-Instruct-2509	FP8	• FP8	• RHAIIS 3.2.5 • RHOAI 3.2
Mistral-Large-3-675B-Instruct-2512	None	• Baseline	• RHAIIS 3.2.5 • RHOAI 3.2
Mistral-Large-3-675B-Instruct-2512-NVFP4	None	• Baseline	• RHAIIS 3.2.5 • RHOAI 3.0
NVIDIA-Nemotron-3-Nano-30B-A3B	FP8	• FP8	• RHAIIS 3.2.5 • RHOAI 3.0

CHAPTER 4. NVFP4 MODELS

The following models, available from RedHat AI on Hugging Face, are validated for use with Red Hat AI Inference.

Table 4.1. NVFP4 Models collection

Model	Quantized variants	Hugging Face model cards	Validated on
Mistral-Large-3-675B-Instruct-2512-NVFP4	None	• Baseline	• RHAIIS 3.2.5 • RHOAI 3.0
Qwen3-VL-235B-A22B-Instruct-NVFP4	None	• Baseline	• RHAIIS 3.3 • RHOAI 3.3

CHAPTER 5. RED HAT AI VALIDATED MODELS - OCTOBER 2025 COLLECTION

The following models, available from RedHat AI on Hugging Face, are validated for use with Red Hat AI Inference.

Table 5.1. Red Hat AI validated models - October 2025 collection

Model	Quantized variants	Hugging Face model card	Validated on
gpt-oss-120b	None	• Baseline	• RHAIIS 3.2.2 • RHOAI 2.25
gpt-oss-20b	None	• Baseline	• RHAIIS 3.2.2 • RHOAI 2.25
NVIDIA-Nemotron-Nano-9B-v2	INT4, FP8	• INT4 • FP8	• RHAIIS 3.2.2 • RHOAI 2.25
Qwen3-Coder-480B-A35B-Instruct	FP8	• FP8	• RHAIIS 3.2.2 • RHOAI 2.25
Voxtral-Mini-3B-2507	FP8	• FP8	• RHAIIS 3.2.2 • RHOAI 2.25
whisper-large-v3-turbo	INT4	• INT4	• RHAIIS 3.2.2 • RHOAI 2.25

CHAPTER 6. VALIDATED MODELS ON HUGGING FACE - SEPTEMBER 2025 COLLECTION

The following models, available from RedHat AI on Hugging Face , are validated for use with Red Hat AI Inference.

Table 6.1. Red Hat AI validated models - September 2025 collection

Model	Quantized variants	Hugging Face model card	Validated on
DeepSeek-R1-0528	INT4	• INT4	• RHAIIS 3.2.1 • RHOAI 2.24
gemma-3n-E4B-it	FP8	• FP8	• RHAIIS 3.2.1 • RHOAI 2.24
Kimi-K2-Instruct	INT4	• INT4	• RHAIIS 3.2.1 • RHOAI 2.24
Qwen3-8B	FP8	• FP8	• RHAIIS 3.2.1 • RHOAI 2.24

CHAPTER 7. VALIDATED MODELS ON HUGGING FACE - MAY 2025 COLLECTION

The following models, available from RedHat AI on Hugging Face , are validated for use with Red Hat AI Inference.

Table 7.1. Red Hat AI validated models - May 2025 collection

Model	Quantized variants	Hugging Face model card	Validated on
gemma-2-9b-it	FP8	• Baseline • FP8	• RHAIIS 3.0 • RHELAI 1.5 • RHOAI 2.20
granite-3.1-8b-base	INT4	• INT4	• RHAIIS 3.0 • RHELAI 1.5 • RHOAI 2.20
granite-3.1-8b-instruct	INT4, INT8, FP8	• Baseline • INT4 • INT8 • FP8	• RHAIIS 3.0 • RHELAI 1.5 • RHOAI 2.20
Llama-3.1-8B-Instruct	None	• Baseline	• RHAIIS 3.0 • RHELAI 1.5 • RHOAI 2.20
Llama-3.1-Nemotron-70B-Instruct-HF	FP8	• Baseline • FP8	• RHAIIS 3.0 • RHELAI 1.5 • RHOAI 2.20

Model	Quantized variants	Hugging Face model card	Validated on
Llama-3.3-70B-Instruct	INT4, INT8, FP8	• Baseline • INT4 • INT8 • FP8	• RHAIIS 3.0 • RHELAI 1.5 • RHOAI 2.20
Llama-4-Maverick-17B-128E-Instruct	FP8	• Baseline • FP8	• RHAIIS 3.0 • RHELAI 1.5 • RHOAI 2.20
Llama-4-Scout-17B-16E-Instruct	INT4, FP8	• Baseline • INT4 • FP8	• RHAIIS 3.0 • RHELAI 1.5 • RHOAI 2.20
Meta-Llama-3.1-8B-Instruct	INT4, INT8, FP8	• INT4 • INT8 • FP8	• RHAIIS 3.0 • RHELAI 1.5 • RHOAI 2.20
Mistral-Small-24B-Instruct-2501	INT4, INT8, FP8	• Baseline • INT4 • INT8 • FP8	• RHAIIS 3.0 • RHELAI 1.5 • RHOAI 2.20
Mistral-Small-3.1-24B-Instruct-2503	INT4, INT8, FP8	• Baseline • INT4 • INT8 • FP8	• RHAIIS 3.0 • RHELAI 1.5 • RHOAI 2.20

Model	Quantized variants	Hugging Face model card	Validated on
Mixtral-8x7B-Instruct-v0.1	None	• Baseline	• RHAIIS 3.0 • RHELAI 1.5 • RHOAI 2.20
phi-4	INT4, INT8, FP8	• Baseline • INT4 • INT8 • FP8	• RHAIIS 3.0 • RHELAI 1.5 • RHOAI 2.20
Qwen2.5-7B-Instruct	INT4, INT8, FP8	• Baseline • INT4 • INT8 • FP8	• RHAIIS 3.0 • RHELAI 1.5 • RHOAI 2.20

CHAPTER 8. VALIDATED OCI ARTIFACT MODEL CONTAINER IMAGES

The following table lists validated OCI artifact model container images available from the Red Hat container registry, including baseline and quantized variants for each supported model.

Table 8.1. Validated OCI artifact model container images

Model	Quantized variants	ModelCar images
llama-4-scout-17b-16e-instruct	INT4, FP8	• Baseline: **registry.redhat.io/rhelai1/llama-4-scout-17b-16e-instruct:1.5** • INT4: **registry.redhat.io/rhelai1/llama-4-scout-17b-16e-instruct-quantized-w4a16:1.5** • FP8: **registry.redhat.io/rhelai1/llama-4-scout-17b-16e-instruct-fp8-dynamic:1.5**
llama-4-maverick-17b-128e-instruct	FP8	• Baseline: **registry.redhat.io/rhelai1/llama-4-maverick-17b-128e-instruct:1.5** • FP8: **registry.redhat.io/rhelai1/llama-4-maverick-17b-128e-instruct-fp8:1.5**
mistral-small-3-1-24b-instruct-2503	INT4, INT8, FP8	• Baseline: **registry.redhat.io/rhelai1/mistral-small-3-1-24b-instruct-2503:1.5** • INT4: **registry.redhat.io/rhelai1/mistral-small-3-1-24b-instruct-2503-quantized-w4a16:1.5** • INT8: **registry.redhat.io/rhelai1/mistral-small-3-1-24b-instruct-2503-quantized-w8a8:1.5** • FP8: **registry.redhat.io/rhelai1/mistral-small-3-1-24b-instruct-2503-fp8-dynamic:1.5**

Model	Quantized variants	ModelCar images
llama-3-3-70b-instruct	INT4, INT8, FP8	• Baseline: **registry.redhat.io/rhelai1/llama-3-3-70b-instruct:1.5** • INT4: **registry.redhat.io/rhelai1/llama-3-3-70b-instruct-quantized-w4a16:1.5** • INT8: **registry.redhat.io/rhelai1/llama-3-3-70b-instruct-quantized-w8a8:1.5** • FP8: **registry.redhat.io/rhelai1/llama-3-3-70b-instruct-fp8-dynamic:1.5**
llama-3-1-8b-instruct	INT4, INT8, FP8	• Baseline: **registry.redhat.io/rhelai1/llama-3-1-8b-instruct:1.5** • INT4: **registry.redhat.io/rhelai1/llama-3-1-8b-instruct-quantized-w4a16:1.5** • INT8: **registry.redhat.io/rhelai1/llama-3-1-8b-instruct-quantized-w8a8:1.5** • FP8: **registry.redhat.io/rhelai1/llama-3-1-8b-instruct-fp8-dynamic:1.5**

Model	Quantized variants	ModelCar images
granite-3-1-8b-instruct	INT4, INT8, FP8	• Baseline: **registry.redhat.io/rhelai1/granite-3-1-8b-instruct:1.5** • INT4: **registry.redhat.io/rhelai1/granite-3-1-8b-instruct-quantized-w4a16:1.5** • INT8: **registry.redhat.io/rhelai1/granite-3-1-8b-instruct-quantized-w8a8:1.5** • FP8: **registry.redhat.io/rhelai1/granite-3-1-8b-instruct-fp8-dynamic:1.5**
phi-4	INT4, INT8, FP8	• Baseline: **registry.redhat.io/rhelai1/phi-4:1.5** • INT4: **registry.redhat.io/rhelai1/phi-4-quantized-w4a16:1.5** • INT8: **registry.redhat.io/rhelai1/phi-4-quantized-w8a8:1.5** • FP8: **registry.redhat.io/rhelai1/phi-4-fp8-dynamic:1.5**

Model	Quantized variants	ModelCar images
qwen2-5-7b-instruct	INT4, INT8, FP8	• Baseline: **registry.redhat.io/rhelai1/qwen2-5-7b-instruct:1.5** • INT4: **registry.redhat.io/rhelai1/qwen2-5-7b-instruct-quantized-w4a16:1.5** • INT8: **registry.redhat.io/rhelai1/qwen2-5-7b-instruct-quantized-w8a8:1.5** • FP8: **registry.redhat.io/rhelai1/qwen2-5-7b-instruct-fp8-dynamic:1.5**
mistral-small-24b-instruct-2501	INT4, INT8, FP8	• Baseline: **registry.redhat.io/rhelai1/mistral-small-24b-instruct-2501:1.5** • INT4: **registry.redhat.io/rhelai1/mistral-small-24b-instruct-2501-quantized-w4a16:1.5** • INT8: **registry.redhat.io/rhelai1/mistral-small-24b-instruct-2501-quantized-w8a8:1.5** • FP8: **registry.redhat.io/rhelai1/mistral-small-24b-instruct-2501-fp8-dynamic:1.5**
mixtral-8x7b-instruct-v0-1	None	• Baseline: **registry.redhat.io/rhelai1/mixtral-8x7b-instruct-v0-1:1.4**
granite-3-1-8b-base	INT4 (baseline currently unavailable)	• INT4: **registry.redhat.io/rhelai1/granite-3-1-8b-base-quantized-w4a16:1.5**

Model	Quantized variants	ModelCar images
granite-3.1-8b-starter-v2	None	• Baseline: **registry.redhat.io/rhelai1/granite-3.1-8b-starter-v2:1.5**
llama-3-1-nemotron-70b-instruct-hf	FP8	• Baseline: **registry.redhat.io/rhelai1/llama-3-1-nemotron-70b-instruct-hf:1.5** • FP8: **registry.redhat.io/rhelai1/llama-3-1-nemotron-70b-instruct-hf-fp8-dynamic:1.5**
gemma-2-9b-it	FP8	• Baseline: **registry.redhat.io/rhelai1/gemma-2-9b-it:1.5** • FP8: **registry.redhat.io/rhelai1/gemma-2-9b-it-fp8:1.5**
deepseek-r1-0528	INT4 (baseline currently unavailable)	• INT4: **registry.redhat.io/rhelai1/deepseek-r1-0528-quantized-w4a16:1.5**
qwen3-8b	FP8 (baseline currently unavailable)	• FP8: **registry.redhat.io/rhelai1/qwen3-8b-fp8-dynamic:1.5**
kimi-k2-instruct	INT4 (baseline currently unavailable)	• INT4: **registry.redhat.io/rhelai1/kimi-k2-instruct-quantized-w4a16:1.5**
gemma-3n-e4b-it	FP8 (baseline currently unavailable)	• FP8: **registry.redhat.io/rhelai1/gemma-3n-e4b-it-fp8-dynamic:1.5**

Model	Quantized variants	ModelCar images
gpt-oss-120b	None	• Baseline: **registry.redhat.io/rhelai1/gpt-oss-120b:1.5**
gpt-oss-20b	None	• Baseline: **registry.redhat.io/rhelai1/gpt-oss-20b:1.5**
qwen3-coder-480b-a35b-instruct	FP8 (baseline currently unavailable)	• FP8: **registry.redhat.io/rhelai1/qwen3-coder-480b-a35b-instruct-fp8:1.5**
whisper-large-v3-turbo	INT4 (baseline currently unavailable)	• INT4: **registry.redhat.io/rhelai1/whisper-large-v3-turbo-quantized-w4a16:1.5**
voxtral-mini-3b-2507	FP8 (baseline currently unavailable)	• FP8: **registry.redhat.io/rhelai1/voxtral-mini-3b-2507-fp8-dynamic:1.5**
nvidia-nemotron-nano-9b-v2	FP8 (baseline currently unavailable)	• FP8: **registry.redhat.io/rhelai1/nvidia-nemotron-nano-9b-v2-fp8-dynamic:1.5**

CHAPTER 9. VALIDATED RED HAT AI MODELCAR CONTAINER IMAGES

Table 9.1. Validated Red Hat AI ModelCar container images

Model	Quantized variants	ModelCar images
llama-4-scout-17b-16e-instruct	INT4, FP8	- Baseline: **registry.redhat.io/rhelai1/modelcar-llama-4-scout-17b-16e-instruct:1.5** - INT4: **registry.redhat.io/rhelai1/modelcar-llama-4-scout-17b-16e-instruct-quantized-w4a16:1.5** - FP8: **registry.redhat.io/rhelai1/modelcar-llama-4-scout-17b-16e-instruct-fp8-dynamic:1.5**
llama-4-maverick-17b-128e-instruct	FP8	- Baseline: **registry.redhat.io/rhelai1/modelcar-llama-4-maverick-17b-128e-instruct:1.5** - FP8: **registry.redhat.io/rhelai1/modelcar-llama-4-maverick-17b-128e-instruct-fp8:1.5**
mistral-small-3-1-24b-instruct-2503	INT4, INT8, FP8	- Baseline: **registry.redhat.io/rhelai1/modelcar-mistral-small-3-1-24b-instruct-2503:1.5** - INT4: **registry.redhat.io/rhelai1/modelcar-mistral-small-3-1-24b-instruct-2503-quantized-w4a16:1.5** - INT8: **registry.redhat.io/rhelai1/modelcar-mistral-small-3-1-24b-instruct-2503-quantized-w8a8:1.5** - FP8: **registry.redhat.io/rhelai1/modelcar-mistral-small-3-1-24b-instruct-2503-fp8-dynamic:1.5**

Model	Quantized variants	ModelCar images
llama-3-3-70b-instruct	INT4, INT8, FP8	• Baseline: **registry.redhat.io/rhelai1/modelcar-llama-3-3-70b-instruct:1.5** • INT4: **registry.redhat.io/rhelai1/modelcar-llama-3-3-70b-instruct-quantized-w4a16:1.5** • INT8: **registry.redhat.io/rhelai1/modelcar-llama-3-3-70b-instruct-quantized-w8a8:1.5** • FP8: **registry.redhat.io/rhelai1/modelcar-llama-3-3-70b-instruct-fp8-dynamic:1.5**
llama-3-1-8b-instruct	INT4, INT8, FP8	• Baseline: **registry.redhat.io/rhelai1/modelcar-llama-3-1-8b-instruct:1.5** • INT4: **registry.redhat.io/rhelai1/modelcar-llama-3-1-8b-instruct-quantized-w4a16:1.5** • INT8: **registry.redhat.io/rhelai1/modelcar-llama-3-1-8b-instruct-quantized-w8a8:1.5** • FP8: **registry.redhat.io/rhelai1/modelcar-llama-3-1-8b-instruct-fp8-dynamic:1.5**

Model	Quantized variants	ModelCar images
granite-3-1-8b-instruct	INT4, INT8, FP8	• Baseline: **registry.redhat.io/rhelai1/modelcar-granite-3-1-8b-instruct:1.5** • INT4: **registry.redhat.io/rhelai1/modelcar-granite-3-1-8b-instruct-quantized-w4a16:1.5** • INT8: **registry.redhat.io/rhelai1/modelcar-granite-3-1-8b-instruct-quantized-w8a8:1.5** • FP8: **registry.redhat.io/rhelai1/modelcar-granite-3-1-8b-instruct-fp8-dynamic:1.5**
phi-4	INT4, INT8, FP8	• Baseline: **registry.redhat.io/rhelai1/modelcar-phi-4:1.5** • INT4: **registry.redhat.io/rhelai1/modelcar-phi-4-quantized-w4a16:1.5** • INT8: **registry.redhat.io/rhelai1/modelcar-phi-4-quantized-w8a8:1.5** • FP8: **registry.redhat.io/rhelai1/modelcar-phi-4-fp8-dynamic:1.5**

Model	Quantized variants	ModelCar images
qwen2-5-7b-instruct	INT4, INT8, FP8	• Baseline: **registry.redhat.io/rhelai1/modelcar-qwen2-5-7b-instruct:1.5** • INT4: **registry.redhat.io/rhelai1/modelcar-qwen2-5-7b-instruct-quantized-w4a16:1.5** • INT8: **registry.redhat.io/rhelai1/modelcar-qwen2-5-7b-instruct-quantized-w8a8:1.5** • FP8: **registry.redhat.io/rhelai1/modelcar-qwen2-5-7b-instruct-fp8-dynamic:1.5**
mistral-small-24b-instruct-2501	INT4, INT8, FP8	• Baseline: **registry.redhat.io/rhelai1/modelcar-mistral-small-24b-instruct-2501:1.5** • INT4: **registry.redhat.io/rhelai1/modelcar-mistral-small-24b-instruct-2501-quantized-w4a16:1.5** • INT8: **registry.redhat.io/rhelai1/modelcar-mistral-small-24b-instruct-2501-quantized-w8a8:1.5** • FP8: **registry.redhat.io/rhelai1/modelcar-mistral-small-24b-instruct-2501-fp8-dynamic:1.5**
mixtral-8x7b-instruct-v0-1	None	• Baseline: **registry.redhat.io/rhelai1/modelcar-mixtral-8x7b-instruct-v0-1:1.4**

Model	Quantized variants	ModelCar images
granite-3-1-8b-base	INT4 (baseline currently unavailable)	• INT4: **registry.redhat.io/rhelai1/modelcar-granite-3-1-8b-base-quantized-w4a16:1.5**
granite-3-1-8b-starter-v2	None	• Baseline: **registry.redhat.io/rhelai1/modelcar-granite-3-1-8b-starter-v2:1.5**
llama-3-1-nemotron-70b-instruct-hf	FP8	• Baseline: **registry.redhat.io/rhelai1/modelcar-llama-3-1-nemotron-70b-instruct-hf:1.5** • FP8: **registry.redhat.io/rhelai1/modelcar-llama-3-1-nemotron-70b-instruct-hf-fp8-dynamic:1.5**
gemma-2-9b-it	FP8	• Baseline: **registry.redhat.io/rhelai1/modelcar-gemma-2-9b-it:1.5** • FP8: **registry.redhat.io/rhelai1/modelcar-gemma-2-9b-it-fp8:1.5**
deepseek-r1-0528	INT4 (baseline currently unavailable)	• INT4: **registry.redhat.io/rhelai1/modelcar-deepseek-r1-0528-quantized-w4a16:1.5**
qwen3-8b	FP8 (baseline currently unavailable)	• FP8: **registry.redhat.io/rhelai1/modelcar-qwen3-8b-fp8-dynamic:1.5**
kimi-k2-instruct	INT4 (baseline currently unavailable)	• INT4: **registry.redhat.io/rhelai1/modelcar-kimi-k2-instruct-quantized-w4a16:1.5**

Model	Quantized variants	ModelCar images
gemma-3n-e4b-it	FP8	• Baseline: **registry.redhat.io/rhelai1/modelcar-gemma-3n-e4b-it:1.5** • FP8: **registry.redhat.io/rhelai1/modelcar-gemma-3n-e4b-it-fp8-dynamic:1.5**
gpt-oss-120b	None	• Baseline: **registry.redhat.io/rhelai1/modelcar-gpt-oss-120b:1.5**
gpt-oss-20b	None	• Baseline: **registry.redhat.io/rhelai1/modelcar-gpt-oss-20b:1.5**
qwen3-coder-480b-a35b-instruct	FP8 (baseline currently unavailable)	• FP8: **registry.redhat.io/rhelai1/modelcar-qwen3-coder-480b-a35b-instruct-fp8:1.5**
whisper-large-v3-turbo	INT4 (baseline currently unavailable)	• INT4: **registry.redhat.io/rhelai1/modelcar-whisper-large-v3-turbo-quantized-w4a16:1.5**
voxtral-mini-3b-2507	FP8 (baseline currently unavailable)	• FP8: **registry.redhat.io/rhelai1/modelcar-voxtral-mini-3b-2507-fp8-dynamic:1.5**
nvidia-nemotron-nano-9b-v2	FP8 (baseline currently unavailable)	• FP8: **registry.redhat.io/rhelai1/modelcar-nvidia-nemotron-nano-9b-v2-fp8-dynamic:1.5**

Model	Quantized variants	ModelCar images
phi-4-reasoning	FP8 (baseline currently unavailable)	• FP8: **registry.redhat.io/rhai/modelcar-phi-4-reasoning-fp8-dynamic:3.0**
qwen3-vl-235b-a22b-instruct-nvfp4	None	• Baseline: **registry.redhat.io/rhai/modelcar-qwen3-vl-235b-a22b-instruct-nvfp4:3.0**
qwen3-next-80b-a3b-instruct	INT4 (baseline currently unavailable)	• INT4: **registry.redhat.io/rhai/modelcar-qwen3-next-80b-a3b-instruct-quantized-w4a16:3.0**
granite-4-0-h-tiny	FP8	• Baseline: **registry.redhat.io/rhai/modelcar-granite-4-0-h-tiny:3.0** • FP8: **registry.redhat.io/rhai/modelcar-granite-4-0-h-tiny-fp8-dynamic:3.0**
granite-4-0-h-small	FP8	• Baseline: **registry.redhat.io/rhai/modelcar-granite-4-0-h-small:3.0** • FP8: **registry.redhat.io/rhai/modelcar-granite-4-0-h-small-fp8-dynamic:3.0**
mistral-large-3-675b-instruct-2512	None	• Baseline: **registry.redhat.io/rhai/modelcar-mistral-large-3-675b-instruct-2512:3.0**

Model	Quantized variants	ModelCar images
mistral-large-3-675b-instruct-2512-nvfp4	None	• Baseline: **registry.redhat.io/rhai/modelcar-mistral-large-3-675b-instruct-2512-nvfp4:3.0**
apertus-8b-instruct-2509	FP8 (baseline currently unavailable)	• FP8: **registry.redhat.io/rhai/modelcar-apertus-8b-instruct-2509-fp8-dynamic:3.0**
nvidia-nemotron-3-nano-30b-a3b	FP8 (baseline currently unavailable)	• FP8: **registry.redhat.io/rhai/modelcar-nvidia-nemotron-3-nano-30b-a3b-fp8:3.0**
ministral-3-14b-instruct-2512	None	• Baseline: **registry.redhat.io/rhai/modelcar-ministral-3-14b-instruct-2512:3.0**

CHAPTER 10. VALIDATED MODELS FOR X86_64 CPU INFERENCE SERVING

The following large language models have been validated for use with Red Hat AI Inference on x86_64 CPUs.

Table 10.1. Validated models for inferencing with x86_64 CPU

Model	Hugging Face model card	Number of parameters
TinyLlama-1.1B-Chat-v1.0	TinyLlama/TinyLlama-1.1B-Chat-v1.0	1.1B
Llama-3.2-1B-Instruct	meta-llama/Llama-3.2-1B-Instruct	1B
granite-3.2-2b-instruct	ibm-granite/granite-3.2-2b-instruct	2B
TinyLlama-1.1B-Chat-v1.0-pruned2.4	RedHatAI/TinyLlama-1.1B-Chat-v1.0-pruned2.4	1.1B (pruned)

IMPORTANT

Quantization formats that require GPU-specific kernels, such as Marlin format, are not supported for CPU inference. Use AWQ or GPTQ quantization formats that are compatible with CPU execution.

The following table provides general guidance for approximate system RAM requirements based on model size:

Table 10.2. Memory requirements for inference serving with x86_64 CPU

Model size	Minimum RAM	Recommended RAM
125M - 500M	8 GB	16 GB
500M - 1B	16 GB	32 GB
1B - 3B	32 GB	64 GB

NOTE

Actual memory usage depends on the model architecture, context length, and batch size. Increase the **VLLM_CPU_KVCACHE_SPACE** environment variable to allocate more memory for the key-value cache when using longer context lengths.

CHAPTER 11. VALIDATED MODELS FOR USE WITH IBM POWER AND IBM SPYRE AI ACCELERATORS

The following large language models are supported for IBM Power systems with IBM Spyre AI accelerators.

NOTE

IBM Spyre AI accelerator cards support FP16 format model weights only. For compatible models, the Red Hat AI Inference inference engine automatically converts weights to FP16 at startup. No additional configuration is needed.

Table 11.1. IBM Granite models for use with IBM Spyre AI accelerators

Model	Hugging Face model card
granite-3.3-8b-instruct	ibm-granite/granite-3.3-8b-instruct
granite-embedding-30m-english	ibm-granite/granite-embedding-30m-english
granite-embedding-107m-multilingual	ibm-granite/granite-embedding-107m-multilingual
granite-embedding-125m-english	ibm-granite/granite-embedding-125m-english
granite-embedding-278m-multilingual	ibm-granite/granite-embedding-278m-multilingual

Table 11.2. Reranker models for use with IBM Spyre AI accelerators

Model	Hugging Face model card
bge-reranker-v2-m3	BAAI/bge-reranker-v2-m3

IMPORTANT

Pre-built IBM Granite models run with the specific Python packages that are included in the Red Hat AI Inference Spyre container image. The models are tied to fixed configurations for Spyre card count, batch size, and input/output context sizes.

Updating or replacing Python packages in the Red Hat AI Inference Spyre container image is not supported.

CHAPTER 12. VALIDATED MODELS FOR USE WITH IBM Z AND IBM SPYRE AI ACCELERATORS

The following large language models are supported for IBM Z systems with IBM Spyre AI accelerators.

NOTE

IBM Spyre AI accelerator cards support FP16 format model weights only. For compatible models, the Red Hat AI Inference inference engine automatically converts weights to FP16 at startup. No additional configuration is needed.

Table 12.1. Decoder models for use with IBM Spyre AI accelerators

Model	Hugging Face model card
granite-3.3-8b-instruct	ibm-granite/granite-3.3-8b-instruct

IMPORTANT

Pre-built IBM Granite models run with the specific Python packages that are included in the Red Hat AI Inference Spyre container image. The models are tied to fixed configurations for Spyre card count, batch size, and input/output context sizes.

Updating or replacing Python packages in the Red Hat AI Inference Spyre container image is not supported.

CHAPTER 13. VALIDATED MODELS FOR GEOSPATIAL INFERENCE WITH TERRATORCH

The following IBM and NASA Prithvi geospatial foundation models are validated for use with AI Inference and TerraTorch.

NOTE

Prithvi-EO-2.0 models use the Vision Transformer (ViT) architecture and require TerraTorch as the model implementation backend. These models accept GeoTIFF imagery as input and return segmentation predictions.

Table 13.1. Prithvi geospatial models for use with TerraTorch

Model	Use case	Hugging Face model card	Validated on
Prithvi-EO-2.0-300M-TL-Sen1Floods11	Flood detection and mapping	Prithvi-EO-2.0-300M-TL-Sen1Floods11	RHAIIS 3.3
Prithvi-EO-2.0-300M-BurnScars	Burn scar detection	Prithvi-EO-2.0-300M-BurnScars	RHAIIS 3.3

Explore the IBM and NASA geospatial models collection on Hugging Face.

IMPORTANT

Prithvi geospatial models are validated for use with NVIDIA CUDA AI accelerators only.

These models require specific vLLM server arguments to function correctly. You must include **--skip-tokenizer-init**, **--enforce-eager**, and **--enable-mm-embeds** when starting the inference server.

For more information, see Serving TerraTorch Models with vLLM.